A SHORT HISTORY OF THE ORCHIL DYES

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The following historical review of orchil dyes represents a collation of the scattered data on this subject. The principal aim has been to survey the information on these particular lichen dyes relating it to their varied usage in the textile trades through the ages. A summary of their methods of manufacture and application is also appended. The term orchil was introduced into England, probably during the fourteenth century, to denote the dye specifically made from several species of *Roccella* that was first manufactured in Florence and known there as oricello. Latterly this name became corrupted to describe all purple and red dyes made from a wide range of lichens, many of which also had vernacular and trade names.

The first section of the paper outlines a general historical survey ; the following two sections deal with the methods of preparing orchil as practised at different times, the purposes for which it was employed, and its application in dyeing.

General Historical Survey of Orchil and Related Dyes

Lichens and Tyrian Purple

The early history of red and purple dyes made from lichens would seem to be inextricably involved with that of Tyrian, or murex, purple, a famous dye which was first extracted from particular species of mollusc by the Phoenicians. The importance of purple in denoting and conferring regality and power in Roman times is well known, and the costliness, both of the dye and the textiles coloured with it, has become legendary. Purple was also associated with red as a symbol of life. It is worth noting that the molluscs, the most important of which were *Thais haemastomma* (*Purpura haemastomma*), *Murex brandaris*, and *M. trunculus* (Born, 1937, pp. 106–110), gave dye hues ranging from blue to red, and the Greek word porphyra and the Roman purpura seem to have covered all the shades capable of being produced from this source. The Greeks regarded purple as a colour of special significance from prehistoric homeric times. Early Persian, Babylonian, and Jewish rulers arrayed themselves, their followers, palaces, and temples with stuffs and tapestries dyed with this symbolic colour. From the discovery of the purple dye contained in molluscs the Phoenicians developed an extensive branch of their trade which was subsequently taken over by the Greeks and Roman economy in turn and which flourished until destroyed by the Arab invaders *c.* A.D. 638.

A verse from the Bible (Book of Ezekiel, chap. 27, v. 7) concerning the riches of the city of Tyre is usually quoted as the earliest reference to the usage of lichen purple made from *Roccella tinctoria*, and other lichens of the same genus.

"Fine linen and broidered work from Egypt was that which thou spreadest forth to by thy sail ; blue and purple from the Isles of Elishah was that which covered thee".

Though the Isles of Elisha have not been satisfactorily identified, there is enough evidence to assume that they lie within the eastern Mediterranean and Aegean area. The city of Tyre, which gave its name to mollusc purple, is supposed to have been the original centre of its manufacture. As the molluscs were common along adjacent coasts, to judge from unearthed heaps of discarded shells, the Phoenicians must have established dispersed centres of purple manufacture in the eastern Mediterranean and Aegean, as well as the coast of North Africa. Species of *Roccella* were also common along the same rocky coasts and there is more than circumstantial evidence to show that the means of obtaining red and purple dyes from them was also known to the Phoenicians, and that they used these dyes in conjunction with mollusc dyes to obtain some of their shades.

Though the colours obtained from *Roccella* dye, when first applied, were said by Theophrastus (see Bancroft, 1813, p. 291) to be even more beautiful than Tyrian purple, unlike mollusc colours, which were noted for their outstanding qualities of brilliance and fastness, lichen purple colours are unfortunately notoriously fugitive, especially when exposed to bright sunlight. While it is conceivable that lichens might have been used alone in domestic, as opposed to commercial, dyeing during this early era, they could not have rivalled the persistence and stability of mollusc colours. Owing to the great demand and the high cost of producing the latter dye—each mollusc shell only contains a minute quantity of colouring matter* —it is most likely that the chief use for lichen dyes at this time lay in the methods of over-dyeing with mollusc colours, the yarn or cloth previously being given a ground colour with another dyestuff. Also used as a gross ground colour, lichen dyes might have been quite valuable to the dyers, being much more easily collected and prepared than the mollusc dye. The colours so based on a lichen ground were known as the conchoid purples (Born, 1937, p. 113). Lichen dyes have a particular natural affinity for animal fibres, their use being restricted to woollen and silk textiles until the nineteenth century. Other commodities which they are capable of colouring are wood, marble, leather, wine, and foods.

Theophrastus (see Bancroft, 1813, p. 291) describes certain plants growing in Crete that were used for dyeing wool. He refers to these plants as alga marina and from his descriptions Miss A. L. Smith (1921, p. 2) identified two lichens, *Roccella tinctoria* and either a species *Usnea* or *Alectoria*. The two latter species, however, are not amongst those from which red and purple dyes can be produced. A plant which answers to the description of a lichen is mentioned by Pliny the Elder (see Forbes, 1956, p. 108) as having been used to provide a ground colour for the mollusc purple. This plant, which he calls phycos thalassium, was found growing on the rocks and coasts of the eastern Mediterranean, and that which grew in Crete was evidently specially commended. Amongst the numerous islands of this area, Crete and Amorgos appear to have been particularly connected with lichen dyeing. In 1717 Pitton de Tournefort (p. 232) records that he found *Roccella tinctoria* growing at Amorgos and considered that this had been used to dye the cloth of ancient Amorgos, which he describes as "a kind of red linens".

During Roman times, Gaetulian purple was renowned for its high quality and was thought by Pliny to have been manufactured from murex. Forbes (1956, p. 119) states, however, that "very few real murex shells were found in the shell

*8,000 snails of *Murex brandaris* yield 1 gram of dye (see Stearn, 1966, p. 237).

heaps of North Africa and Mogodor Islands", thus proving, he concludes, that Gaetulian purple was chiefly obtained from lichens. In spite of this it seems likely that *Thais haemastomma*, which abounds on the Atlantic coasts of Africa (Born, 1937, p. 107), was the chief ingredient of Gaetulian purple. Lichen dyes would no doubt have been useful in helping to "blue" the natural red colour obtained from *Thais haemastomma*. The Gaetulian purple industry was founded by Juba, King of Mauretania, who, according to Pliny, also made an expedition to the Canary Islands about A.D. 40, where he established dye-works. In the fifteenth or sixteenth century, on the rediscovery of these islands by the Spanish, species of *Roccella* were found to be growing there in great profusion, especially on the sea cliffs. A recently published essay by Oskar Kokoshka concerning a Phoenician silver coin once in his possession is also interesting in connection with early dyeing (Kokoshka, 1962, p. 141). On the reverse of this coin was depicted a plant which the author described as resembling a seaweed and refers to as such throughout the essay. He claimed to have discovered that the secret of the Phoenician purple dye lay, not in mollusc dyes at all, but in this seaweed, and that the ancient dyers had succeeded in deceiving Pliny and other historians as to its true origin. Although seaweeds have been used in dyeing in other capacities, they do not contain any colouring matter capable of dyeing textile fibres. It seems quite possible that the plant represented is intended to be a lichen of the genus *Roccella*, which could appear somewhat similar to a seaweed when depicted on such a small scale and format. The murex shell, possibly as a symbol of a valuable article of commerce, appears on some Phoenician (Huxley, 1954, pp. 73–75) and Roman coins; if the plant on Kokoshka's coin is intended for a lichen, this might indicate its importance in the ancient purple dye industry.

Evidence of lichen dyes being used with other dyes made from plants is also found in a document known as the Stockholm Papyrus, a demotic Greek work dating from the third century A.D. This work, obtained from a Theban tomb in 1928, is now in the Victoria Museum, Uppsala (see Faber, 1938, p. 287). The Stockholm Papyrus, which appears to be the sequel to another treatise called the Leyden Papyrus, is supposed to be based on ancient Egyptian writings. Of the seventy dye recipes given, a great many are concerned with the imitation of mollusc purple colours, apparently with such success that it was thought impossible to tell the difference (see Born, 1937, pp. 113–114). The Stockholm Papyrus also contains what is probably the earliest surviving recipe for obtaining a blue colour by using woad and lichen purple (see Faber, 1938, pp. 289–290). This practice was performed with woad and also with indigo in western Europe up to the nineteenth century and was based on an initial blue grounding which, on further dyeing with lichen purple colour, gave the blue a richer and redder cast and a greater brilliance.

Although well established over all the Mediterranean area by the time of the Roman Empire, the purple industry appears to have mysteriously declined during the latter years of that epoch. True, some of the early Christians denounced the use of purple clothing as a sinful luxury, and this may, in part, have accounted for its disappearance from some of the areas where it was formerly manufactured. And yet the one place the industry was carried on through the Dark Ages until early Medieval times was at Constantinople, the heart of early Christendom for many years. Forbes (1956, p. 119) suggests as an explanation of the disappearance of the lichen dye of Gaetulia the fact that urine was used in its preparation, and this,

together with the nauseous smell in preparation (Jackson, 1917, pp. 1–29), would have been considered unclean by the Arab conquerors of North Africa. But urine was also used in the preparation of murex purple, and this evidently did not prevent the Saracen textile workers of Sicily from taking over the production of the purple dye from Byzantium during the eleventh century A.D. when the Crusader conquest of Constantinople temporarily disrupted the purple industry there (see Born, 1937, p. 122). The Christian Emperors and Princes of the Church kept up a continual demand for purple because of its regal and sacred significance, which had been transferred from pagan ceremonies to the Christian Church, as well as for reasons of personal luxury. Difficulties in obtaining supplies from the production areas now conquered by the Arabs probably raised the prices to still more prohibitive levels and thus led to a concentration on improving the use of the more readily available and comparatively much cheaper kermes dye. This is a red dye from the insect *Kermes ilicis* (L.), related to cochineal (from *Dactylopius coccus* Costa), and lac (from *Laccifer lucca* Kerr), which was much used by the people of the Near East and the Mediterranean in Roman times and before. It is significant that ten years after the final defeat of Constantinople in 1453, the imperial purple of the cardinals' robes was changed by Pope Paul II to a scarlet colour, which was called Cardinal's purple and probably derived from kermes (Born, 1937, p. 123).

Gaul and the Dark Ages

It is usually thought that, with the passing of the influence of Rome in Western Europe, the knowledge of how to prepare dye using *Roccella tinctoria* had been totally forgotten, but there are some indications which suggest that this may not have been entirely so. During the Roman era textile manufacture became established in some districts of Italy and Gaul, and after the fall of the Empire, in Gaul at any rate, these activities were not necessarily disrupted. In fact some of the textile centres which later became famous in France and the Netherlands are said to have originated or derived from these early cloth workers. It also appears that during this period fulling* and dyeing were separated from the total process of textile production and became self-supporting industries (Toutain, 1930, pp. 296–298). Marseilles and Nîmes were cloth centres of the Mediterranean area ; in the north, Rouen, Amiens, Rheims and Bourges were manufacturing textiles of different kinds. In Britain, the Anglo-Saxons were known for their love of fine garments and their skill in embroidery, and they used wall-hangings of purple and other colours in their halls (Digby, 1957, pp. 37–55).

With the fall of the Western Empire the demand for rich garments and tapestries in western Europe certainly did not cease. On the contrary, the breaking up of the Empire amongst numerous chieftains and kingdoms led to an increase in those aspiring to the purple. So, although the most prized colour was still the mollusc purple, now available only from Constantinople, the manufacturers and domestic dyers of the west must certainly have made efforts to produce these colours in other ways. Plinius (ed. Bostock and Riley, pp. 390–391) comments that Transalpine Gaul could produce with the aid of vegetable dyes Tyrian purple, oyster purple and other colours. But these dyes evidently were not fast and were liable to alteration by washing (Louandre, 1859, p. 10). Quite probably the blue of woad (*Isatis*

*Fulling is one of the finishing processes in woollen cloth-making whereby the cloth is shrunk by agitation in a solution of soap and water, either manually or by machine.

tinctoria), and crimson of kermes formed the basis of many of these colours. Although these two dyes were not particularly fugitive, some of the other dyes very likely used with them in order to achieve the various shades and lustre of the mollusc purple were probably much less fast. It is tempting here to imagine that the use of the lichen purples and reds was continued, for a while at any rate, in some parts of Gaul. These colours could certainly be a clue to the methods of imitating the mollusc colours as already seen from the Stockholm Papyrus. They are also fugitive and are altered by alkalis and acids in the process of washing. Until the introduction of chemical colours in the nineteenth century the lichen purple dyes were greatly prized by dyers. No substitute was found for their brilliance and depth of colour. It is not only the shade but the quality of the colour which seems to be unique ; these dyes actually impart a softness and lustre both to silk and wool, whereas many dyes from other sources and their associated processes of mordanting* are harmful to these fibres and have to be used with the greatest care so as not to produce harsh cloth and lustreless colours.

Perhaps the "disappearance" of the *Roccella* purple at this time indicates a cessation of its importance as an article of commerce owing to a general decline in Roman trade in the Mediterranean. However, lichens from local sources would still have been available to some of the existing textile centres even if there was no extensive trade in them. Marseilles and Nîmes could have collected *Roccella* in bulk from the adjacent coasts, as well as species of *Lecanora* and other lichens which give purple and red dyes, from the Alpes Maritimes and the Plateau Centrale. In the north of France and southern Britain lichen dyes may not have been known, but in Scandinavia, western and northern Britain, and Ireland many species of lichens which give red and purple dyes must have been profusely at hand and, according to a domestic rather than commercial tradition, have been used in these parts from early times, particularly by the Gaelic- and Celtic-speaking peoples. Lacmus, a red lichen dye made in Norway and known to have been exported to Germany and England by the early fourteenth century (Leix, 1937, p. 21) was, presumably, a development of these primitive dyes. Similarly the method of producing violet dye from other species of lichen which became an important industry in central France during the eighteenth century must surely have been derived from traditional practice handed down by word of mouth amongst the early communities of those parts.

The Introduction of Oricello to Western Europe c. 1300

However conjectural, due to lack of documentation, the above information may be, there is no doubt that the re-establishment of *Roccella* purple in European commercial trade dates from the early fourteenth century when a Florentine merchant named Ferro, or Federigo, began to manufacture the dye in Florence. He discovered the process of making the dye when voyaging in the Greek islands and the Levant and was successful in keeping the secret of its preparation in Florence for some years. *Roccella tinctoria* was found to be the best for making the colour, but *R. fuciformis* and other species in the genus were also used by the

*Mordants are mineral salts, such as alum, used as agents to fix dyes to textile fibres. The lichen dyes need no mordants, though they are sometimes used to alter the colours. Primitive peoples use plants and naturally occurring substances such as mud, which contain these salts.

Florentines in the manufacture. At what date the distinction between the different species was recognized by dyers is not clear. The dye was called "oricello" and the family of Federigo took the name of Oricellari, later corrupted to Rucellarii or Rucellai. They collected all available supplies of the lichens then known from the Levant, the coasts of the Aegean and of the Mediterranean and held a monopoly for manufacturing oricello for about a hundred years. The much-quoted account of how the *Roccella* dye was brought from the Levant was written by G. Marcotti and privately printed in Florence in 1881 (Woodward, 1949, pp. 302–307).

Trade in the dyestuff apparently extended to most of the textile centres of Europe. In the late fourteenth century Francesco di Marco Datini, a merchant of Tuscany, was importing the oricello lichens from Majorca and exporting the dyestuff to the Spanish markets, together with other dyes and ingredients for dyeing of Italian, Balkan and Near Eastern origin (Origo, 1963, pp. 69, 97). Some time during the fifteenth or sixteenth century supplies of the lichens were also found to be available in the Canary Islands, which were conquered for Spain by the Norman captains, Gandifer de la Salle and John de Bethencourt, about 1404 and became legal property of the Castilian crown in 1476 (Anon. (1), 1929, p. 729).

The Naming of Orchil Dyes and Lichens in Western Europe

In Spain the dye was known as orcigilia or orciglio, orchilla or orchillo; in Portugal, urzela or urzella; in France, orseille, orchal and orselle; in Germany, persis; in England, orchal, archil, orchil, and sometimes orchella or orchilla. Possible derivations have been explained by Carol H. Woodward (1949, pp. 302–307) in a paper entitled "Vernacular names for *Roccella*", who concludes that the name adopted for the dye and the Rucellai family derives from an unidentified vernacular name of the lichen, rather than *vice versa* as has sometimes been suggested. Other names are also found; Guicciardini writing in French in 1560 (see Tawney and Power, 1951, p. 169) uses orcille, and Rees' Cyclopoedia (1819) enumerates argol as one of the names for archil, though this name does not appear to have been used elsewhere. Argol or argal, which was a form of tartar (bismuth) used in dyeing, was possibly a cause of confusion in this case.

It seems that these names were generally applied to the lichens as well as to the dye made from them, although roccella and raspe or raspa were specifically used by the Florentines for the lichens. Frequently it is difficult to decide whether the plant or the dye is being referred to. Later, during the eighteenth century and onwards, the roccella lichens were known in commerce as weeds, and are so called today. Those collected in the Canary Islands were called canary weed, and so on. Dye made from *Roccella* was often called herb archil, orseille de l'herbe. In English, orchil and archil seem to have been in most common usage, and all further references to this dyestuff in this paper are cited as orchil. By the end of the nineteenth century orchil had come to mean any purple or red dye made from lichens.

The End of the Florentine Monopoly

A clearer distinction in the matter of naming could assist in discovering more precisely when the monopoly of manufacture passed from Florentine hands. According to Leggett (1944, p. 58) an Italian cloth dyer in London was the first to reveal the secret, some time after the discovery of orchil lichens in the Canary

Islands, *c.* 1703. However, C. J. Cooksey, in an article on "Litmus and Related Pigments" (1964, pp. 5–9), points out that there are references to "Orchell or Cork called Jarecork" and "The Crafte to make orchell" in Arnold's Chronicle of 1502. The use of the words "crafte to make" indicates that the method of making orchil was no longer a Florentine secret. Cork is a crimson dye made in the Highlands of Scotland, known locally as corcur or korkir, and similar to the already mentioned lacmus. It appears from this that the similarity between orchil and the Scottish corcur was appreciated. Orchall is mentioned in a Statute of Henry VIII dated 1533 (see Salzmann, 1923, p. 212), together with corke. In his "Plictho de l'Arte de Tentori", published in 1540, Giovanni Ventura Roseto gives the method of making orchil and several recipes for using it. Guicciardini (see Tawney and Power, 1951, p. 169) says in his description of the trade of Antwerp, 1560, "l'Espagne nous éslargiste inombrables éspèces de marchandises . . . de l'orcille de Canaries ; laquelle herbe à taindre est nommé Raspe par les Florentins". It seems from this allusion to raspe that the lichens themselves were being brought to Antwerp, presumably for sale to manufacturers in England and Flanders. Portugal also supplied orcille "from their own country" to Antwerp. Although the Cape Verde Islands had been part of the Portuguese royal dominions since 1495, the presence of species of *Roccella* growing on their rocks and coasts was apparently not recognized until the early nineteenth century.

Imperato in "Dell' Historia naturale", 1599, and John Ray both mention orchil, and the latter quotes a method of preparation from Imperato (Ray, 1686, p. 74). In the "Instruction Générale pour la teinture des laines", 1669, Colbert gives details of the types of work for which the use of orchil was permitted (see Amoreux, 1787, p. 81). Otherwise the seventeenth century seems to lack written evidence of the demand and use for orchil. Nevertheless, it seems likely that the Spanish and Portuguese traders who superseded the Venetians in supplying dyestuffs to the European markets were on the look-out for new sources of lichens to satisfy the growing markets in western Europe. These they discovered during the next hundred years in South America, Morocco, Madeira, the Azores, Sardinia, Angola and Madagascar. By the early eighteenth century there was "a great demand" for roccella weed for London, Amsterdam and Marseilles as well as for Italy (Leggett, 1949, pp. 107–110). In 1717, Pitton de Tournefort found that *Roccella* lichens were sent to England and Alexandria from the Island of Amorgos at ten écus* a quintal†. He also notes that, although the lichens were not scarce on other islands of the Archipelago, it was only at Amorgos that their use as a dye was known (Pitton de Tournefort, 1717, p. 232).

Growth of Trade in Orchil Lichens in the Eighteenth Century

In 1730 the captain of an English ship discovered *Roccella* growing in the Cape Verde Islands and recognised it as a possible valuable cargo. He took a sample back to Santa Cruz and showed it to some Spanish and Genoese merchants. Between them they fitted out a ship in 1731 and took eight Spanish orchil-gatherers to Cape Verde ; in a few days they gathered from St. Anthony and St. Vincent Islands a cargo of 500 quintals, for which they paid the Governor one piastre a quintal. The current price for canary weed before 1725 was three or four piastres

*A French silver crown piece.
†100 kilograms.

a quintal ; in 1730 it was sold in London for £4 a quintal. The Cape Verde *Roccella* was found to be longer and thicker than the canary weed ; six to eight inches long and evidently richer in colouring matter. It was thought that this was because it had not been gathered previously (Hellot *et al.*, 1789, p. 232 ; Bancroft, 1813, p. 294).

Hellot gives a list of amounts exported "one year with another" from the Canary Islands *c.* 1731 : Isle of Fer, 800 quintals ; Teneriffe, 500 ; Canaries, 400 ; Fuerta Ventura, Lansarotto and Gomere, 300. Fifteen to twenty reals per quintal was paid to the gatherers (five shillings per quintal according to Bancroft). The gathering was a difficult and presumably dangerous occupation on account of the nature of the rocks and cliffs on which the *Roccella* grew. These were often very steep, so that the gatherers had to be suspended by ropes over the sea. In Angola it was noted that species of this genus were found growing on trees (Anon. (2), 1848, p. 138).

Expansion of trade with the East and the growth of the textile industries in northern Europe during the eighteenth century, together with the general development in scientific ideas at this time, made it possible for manufacturers to keep pace with the growing demand for silk and cotton goods, as well as for the traditional woollens and linens. The new textiles required new dyes and dyeing techniques and there appears to have been at this period a particularly close liaison between chemists, botanists and manufacturers. This was especially so in France, where the Government recognised the importance of research in this field, which was frequently carried on in the universities, particularly at Lyons and Montpellier. Towards the end of the century comprehensive books on dyeing, giving information on theory, methods and recipes, were being published in most European countries and there was a spate of publications on the economic uses of indigenous plants.

In 1785 the Académie des Sciences, Belles-Lettres et Arts at Lyons offered three prizes for works on lichens and their application in medicine and the arts. Particular attention was to be given to fresh research on the properties of these plants. The three winning memoirs were published together at Lyons in 1787 and were by :

G. F. Hoffmann, doctor of medicine of the University of Erlangen.

Monsieur Amoreux, fils, doctor of medicine in the University of Montpellier.

Monsieur Willemet, demonstrator of botany at Nancy.

Hoffmann's memoir, entitled "Commentatio de vario Lichenum usu", includes painted patterns of colours obtained from lichens and the names used for these colours by the French dyers. That of Amoreux contains a very clear account of the use of orchil in the French dyeing industry. Monsieur Willemet's memoir is less full, but gives some information of the use of lichen dyes among peasants of various countries.

An important dye manual of the time was J. P. Hellot's "The Art of Dyeing Wool", published in Paris in 1750. It was translated into English in 1785 (later editions 1789, 1790, 1901), and published in London in a single volume containing sections on silk dyeing by J. Macquer and cotton dyeing by Le Pileur d'Apligny. P. J. Macquer and C. L. Berthollet (1791) were chemists who did important work on dyes and dyeing at this time. The works of C. L. and A. B. Berthollet continued to be edited and published until 1841 (Berthollet, C. L., 1791 ; Berthollet, C. L. and

A. B., 1791, other editions 1792, 1804, 1824 and 1841). J. P. Westring of Copenhagen carried out experiments on the dyeing properties of Scandinavian lichens and published a memoir on the subject in 1792 (Westring, 1792, pp. 267–269). In 1805–9 he published his "Svenska Lafvarnas Farghistoria" which includes painted examples of the colours he obtained, and contains also very beautifully engraved plates illustrating the lichens. In England, the most important work on the subject of dyeing was Edward Bancroft's "Experimental Researches concerning the Philosophy of Permanent Colours", published in 1813 and containing a comprehensive chapter on the history of lichen dyes (Bancroft, 1813, pp. 291–306).

Parelle, a Lichen Dye Industry Native to France

During the eighteenth century an industry grew to importance in the district of the Auvergne in France for the manufacture of parelle, or orseille d'Auvergne, orseille de terre. This was a purple dye similar to orchil, but made from different lichens. For purposes of distinction perelle or la perelle here refers to the lichen and parelle to the resultant dyestuff. There is some doubt as to which species was the one known locally as la perelle and has, according to Bancroft, been confused with *Ochrolechia parella*. He quotes a Monsieur Cocq (1812, p. 261) as stating that la perelle was really *Variolaria orcina* Ach. (Syn. *Pertusaria corallina* (L.) Arnold, BM Isotype). This species is known in Scotland as white crottle, and was one of the lichens employed there for red and purple dyes. According to Westring and Cocq, it gives a beautiful aramante colour. *Lichen corallinus*, mentioned by Cocq (1812, p. 260) and known locally as barbefine, was not the true perelle. Westring (1792) suggested that la perelle was *Lichen impressus* (*Diploschistes scruposus* var. *bryophilus*). According to M. E. Hale, quoted by E. M. Bolton (1960, p. 58), all the *Diploschistes* species contain the dye principle lecanoric acid. The local French name for *Ochrolechia parella* was la pommelée and this was "constantly rejected by the persons employed to gather la perelle as being unfit for their purpose" (Bancroft, 1813, p. 296). Both Cocq and Westring obtained only poor colours from *Ochrolechia parella*; Cocq a reddish fawn and Westring dark grey. In her experiments Miss Eileen Bolton (1960, p. 48) found this lichen difficult to macerate and only produced a dusky pink. Nevertheless, most writers of the eighteenth and early nineteenth centuries, when the industry flourished, refer to *Lichen parellus* (=*Ochrolechia parella*) as the perelle lichen. At one time la perelle was thought to have been a kind of calcinated earth, which may have given rise to its name of orseille de terre. The name perelle comes from péré, the Languedocien word for rock (Bruslons, 1750, p. 129).

Two sorts of perelle were evidently gathered, white and grey, the difference in colour being supposedly due to the maturity and purity of the plant, though it is possible that more than one species was involved (see Amoreux, 1787, p. 67). The grey lichen was preferred to the white, on the assumption that it was more fully developed (from three to six years' growth) and contained a higher proportion of colouring matter. This was called perelle maîtresse (Duchartre, 1845, pp. 515–516; Cocq, 1812, pp. 258–277). Younger growth was called pucelle, but it is not clear whether this was synonymous with white perelle. The perelle which grew upon granite rocks was supposed to give the finest colour; it was called varenne, but was rare in the Auvergne. *Lichen impressus* (*Diploschistes scruposus* var. *bryophilus*) also had two names, la grise blanche and la grise noire, denoting different stages

of development. Whichever was the species forming the main ingredient of parelle, Duchartre (1848, pp. 515–516) states that it was nearly always mixed with other lichens not having the same tinctorial properties. This may have been partly because of the difficulty of recognising the correct species, or perhaps for fraudulent reasons in order to make bulk. Parelle was inferior to orchil made from *Roccella*, partly because this arbitrary mixture of lichens used in making it rendered the colour less brilliant and unpredictable. It was produced in great quantity and manufacturers sometimes fraudulently tried to pass it off as orseille de l'herbe (orchil).

The instruments used to scrape the lichens from the rocks were small flexible iron blades about the thickness of a knife. These seem to have been made exclusively at St. Flour. The lichens were collected in small pouches. One side of the opening of the pouch was stiffened with a slightly curved iron plate. To either end of the iron plate was attached a wooden semi-circle which held open the mouth of the pouch. The iron plate was held against the rock and the lichens scraped into the pouch. Men, women and children gathered the lichens in the winter during wet weather. Two kilos a day could be collected by the best workers. In 1789 perelle sold for two sols a kilo ; by 1812 the price had doubled. The purchasers tested the quality of the perelle by moistening a small quantity with urine in a glass phial, the correct perelle lichens turned brown, others turned yellow.

Another more persistent use for perelle was in the manufacture of litmus in The Netherlands.

Cudbear—an Orchil Dye made from Lichens Native to the British Isles and N. Europe

In Scotland a similar dyestuff began to be manufactured in the second half of the eighteenth century. While mending a copper boiler in a London dye-house, George Gordon, a coppersmith from Banffshire, noticed the similarity between the orchil dye used there and the native Highland red dyes made from crottles. His brother Cuthbert, a merchant of Leith, was already marketing a purple dye, the preparation of which "from a species of moss" he had himself discovered (A. Clow and N. L. Clow, 1952, p. 210). In 1758 the brother applied for a patent on a lichen dye they called Cudbear, after their mother, whose maiden name had been Cuthbert (Patent Office Library, A.D. 1758, No. 727). They pleaded that the dye was "intirely composed of Materials the Produce of Great Britain or of His Majesty's Plantations". The names of the ingredients are given as :

(1) "Lichen, being a vegetable that grows on very large stones".

(2) "Muscus rupitus admiscens, or coroloides, that mostly grows on shattered rocks or very large stones where it can have a small portion of earth".

(3) "Muscus pixydatus, being a weed, plant or vegetable that grows in low, moorish, tufty grounds".

The chief lichen used for making cudbear was *Ochrolechia tartarea* (including *O. androgyna*), which yields a very rich purple colour almost as brilliant as orchil (Calvert, 1876, p. 247 ; Ure, 1839, p. 377). *Urceolaria calcarea* (=*Lecanora calcarea*) was another ingredient, and *Cladonia pyxidata*, which can be recognised as "Muscus pyxidatus". Doubtless, as in the case of parelle, other species were sometimes involved. Sir James Edward Smith comments in a note with a specimen

of *O. tartarea* in his herbarium (sheet No. 1693.9), now in the Linnean Society of London, that "the gatherers are very accurate in taking only such specimens as are of the firmest texture, and they never scrape the same rock oftener than once in five years". Pennant records that "corcur or the Lichen omphalodes" was "exported" from the Earl of Breadalbane's estate at Taymouth for the use of dyers at one shilling or one shilling and fourpence a stone (Pennant, 1772, p. 36). The gatherers used iron hoops for scraping the rocks and gathering took place generally in August on the assumption that the summer sun enriched the dyeing potential of the lichens used.

The Gordon brothers went into partnership with Messrs. William Alexander and Co. of Edinburgh, who started to manufacture the dye in 1758. George Gordon died in 1764 and Cuthbert was then joined by two younger brothers, James and William, but the company failed in 1778. In the meantime a factory had been started in Glasgow for making cudbear, with James Gordon as one of the partners. It was under the management of George Macintosh, father of the inventor of waterproofed garments, who had already been successfully involved in the manufacture of various chemical products connected with the dyeing industry. This company proved to be the most successful in producing cudbear, buying up the original Edinburgh business after its failure and continuing in operation until 1852.

In spite of attempts to keep the method of preparation secret, including the building of 10 ft. walls around the factories and the employment of Highlanders speaking only the Gaelic language, information leaked out that led to the establishment of a cudbear works in Great Peter Street, Westminster, London, in 1793. This company did not last for more than ten years. Attempts to manufacture cudbear in Liverpool and Ireland were also unsuccessful. Archibald and Nan Clow, whose historical account of the cudbear manufacture is possibly the fullest yet published, suggest that Macintosh's success was due to his scientific method of production (Clow and Clow, 1952, p. 213).

The lichens required by the cudbear manufacturers were at first obtained from Scottish sources, the Shetland Islands, Westmorland, Cumberland and Derbyshire. Later they were imported in large amounts from Norway and Sweden. Bancroft (1813, p. 381) states that twenty to thirty pounds a day could be collected. One penny a pound was paid in Westmorland and Cumberland (Brewster, 1830); at Fort Augustus three shillings and fourpence per twenty-two pounds (J. E. Smith, 1833, p. 191). Imported lichens sold at £20 to £25 per ton, having risen from £3, the price when they were first in demand (Clow and Clow, 1952, p. 211). As much as £45 a ton was paid during the Napoleonic Wars. At the same time *Roccella* was selling for as much as £300 a ton, but in addition to a lower concentration of dye principle there was more waste material in cudbear lichens, and by the time they had been washed and dried ready for use they had been reduced in weight by half. The cudbear and parelle lichens were known commercially as "moss" to distinguish them from the orchil "weed". In 1823 *Umbilicaria pustulata* was found to be a good substitute for *O. tartarea* (Clow and Clow, 1952, p. 211), and from personal observation this species makes an excellent deep purple.

In an endeavour to revive the occupation of lichen harvesting in the Scottish Highlands for the benefit of that under-developed community, Dr. Lindsay (1854, pp. 228–250) made a series of investigations and experiments to show that a number

of native lichens could be used economically to produce cudbear. Among those which he listed were species of *Roccella, Lecanora, Umbilicaria, Gyrophora, Urceolaria, Parmelia, Evernia, Borrera, Ramalina, Lecidea, Isidium,* and *Sphaerophorus.* But the commercial use of native lichens was never successfully revived and before long the dye was being manufactured from species of *Roccella* and the name thus became synonymous with orchil.

Continued Use of Lichen Dyes in the Nineteenth Century

In spite of the rapid development of chemical dyes after the discovery of coal tar colours in 1858 there continued to be a reduced market for orchil and cudbear. The quality of the lichens coming from the Canaries and Cape Verde declined, due no doubt to over-harvesting. A new source was found for England in India and Ceylon. A species identified as *Roccella montagnei* was "extremely common" in India, growing on the trunks of mango trees (Anon. (2), 1848, p. 138). Other supplies came from Mozambique, Angola, Peru, Madagascar and Zanzibar, imported by the Portuguese, English and French. Prices rose to £300 per ton for roccella weed in London in 1813; sometimes up to £1,000 per ton according to Bancroft (1813, p. 294). In 1851 "*Roccella fuciformis*" from Ceylon fetched £380 a ton and "*Parmelia perlata*" £190 to £225 a ton. But by 1867 prices had dropped considerably to between £70 and £30 a ton; "*Roccella fuciformis*" from Mozambique £50, and from Ceylon, £20 to £30. Lindsay (1867, pp. 101–109) notes that the higher prices commanded by the trade in the early nineteenth century reflect an increase in the demand for lichens and that the manufacture of orchil and cudbear was "flourishing more than ever". It is quite possible that at this time the Dutch litmus trade was also expanding at a greater rate than before and was contributing considerably to this rise in the demand for lichens.

By 1867 the orchil manufacturers of London had absorbed the cudbear industry and the two dyestuffs became analogous, being made by this time from the same species of lichen. The principal use for the dye appeared to be in the manufacture of carpets and for staining leather. Cudbear was still in use domestically in the Hebrides, Orkneys and Shetlands (Lindsay, 1867, pp. 101–109) and the Republic of Ireland (Aran and Blasquet Islands).

There are occasional references to the use of orchil in textile printing during the early nineteenth century. This was evidently for printing woollen and silk goods and for mixed cloths such as mousseline de laine (often wool and cotton) and challis (silk and wool) (Ure, 1839, p. 236).

An important development in 1856 was the discovery by Monsieur Marnas of Lyons of a method of producing fast orchil purple and mauve dyes. These were made under the name of French purple and used for dyeing woollen and silks as well as cottons. An example of silk dyed with French purple is included by F. C. Calvert in his "Dyeing and Calico Printing" (1876). This book, edited by John Stenhouse and C. E. Groves, includes a chapter on orchil which is interesting because it is probably one of the last manuals to be published on the subject at a time when the dye was still in fairly common use. It is strange that, after years of searching on the part of dyers for a way to make the orchil dyes more durable, French purple, which evidently was fast, seems not to have survived as long as the original orchil or cudbear.

Orchil Dyes in the Twentieth Century

In the early part of this century orchil was still used in silk and woollen dyeing, but there is no evidence of its being employed in the textile industry to-day. Although the 1962 edition of the Encyclopaedia Britannica indicated that orchil was still in use for carpet yarns, the account given in the second edition of the Colour Index, 1956, makes no mention of any modern use for the dye. The Wilton Royal Carpet Factory has informed me that natural dyes have not been used by them for over forty years.

There appears, nevertheless, to have been a revival of the orchil manufacture in this country occasioned by the cessation of supplies of litmus from The Netherlands in the Second World War. When Messrs. Johnsons of Hendon, makers of litmus paper, were unable to obtain litmus from The Netherlands on account of the war they decided to start making it in their own works. After considerable research into the methods of production and the possibility of obtaining the necessary lichens, they were able to start production. The enterprise grew to include the production of orchil, which they call cudbear, and orcinol.

Two lichens are imported; *Roccella tinctoria* from the Cape Verde Islands for making litmus and *Roccella montagnei* from Madagascar for cudbear. The latter contains the greatest amount of colouring matter, from three to four per cent by weight.

The cudbear is sold in this country chiefly to food manufacturers as a colouring agent for soft drinks, jams, and other preserves. For this purpose the lichen dye probably has a certain attraction at the present time, in view of the suspected adverse affects on health of some synthetic colourings used in foods. Most of the cudbear produced by Messrs. Johnsons of Hendon is exported to America, but for what purpose they do not know. According to Llano (1951, pp. 385–422), quantities of lichens are also imported into the United States.

Another chemical firm in the Midlands produces orcein, the active principal found in orchil dyes, from *Roccella fuciformis* and *Roccella tinctoria*. Their orcein is used in stains for microscopic preparations.

It seems therefore that, although they are no longer employed in the textile trade, the red and purple lichen dyes may still have a useful function in other fields.

Methods of Preparing Orchil, French Purple, Parelle and Cudbear

Briefly, the preparation of the purple dye from lichens consists in the extraction of unique and particular colour-yielding depsides, such as gyrophoric, lecanoric, evernic acids and erythrin, which on decomposition give, in dyeing terminology, orcin, or its derivatives, the colour-yielding principle (Edge, 1914, p. 187). This is converted by oxygen into orcein, the actual colouring matter present in orchil. It appears that a single ammoniacal maceration and fermentation achieves the first two stages of the process, whilst the introduction of air by stirring completes the operation. The most important orchil-yielding substances are:

Erythrin ($C_{20}H_{22}O_{11}$)—found in most species of *Roccella*, including *R. fuciformis*, *R. montagnei*, and *R. peruensis*.

Lecanoric acid ($C_{16}H_{14}O_{11}$)—found in *Roccella tinctoria*, *R. portentosa*, *R. sinuensis*; all *Diploschistes* species; *Parmelia glabratula* and several other *Parmelia* species, etc.

Evernic acid ($C_{17}H_{16}O_7$)—*Evernia prunastri*; *Ramalina pollinaria*.

Gyrophoric acid ($C_{24}H_{20}O_{11}$)—found in *Umbilicaria* species; *Ochrolechia tartarea*, *O. frigida*, and *O. androgyna*; *Lobaria pulmonaria*, etc.

(See Asahina and Shibata, 1954.)

Calvert (1876, p. 233) comments that Robiquet first detected orcin in *Variolaria dealbata* (*Pertusaria corallina* (L.) Arnold) in 1829.

Orchil from Roccella

The earliest known descriptions of the preparation of orchil are given by Roseto (1540), Imperato (1599, p. 735) and Michele (1729, p. 77). The method described by Roseto and quoted by Hellot is as follows:

"Take one pound of the Orselle of the Levant, very clean; moisten it with a little urine; add to this sal-ammoniac, sal-gemmae ("sel terrestre et fossile", des Bruslons, p. 184), and saltpetre, of each two ounces; pound them well, mix them together, and let them remain so during twelve days, stirring them twice a day; and then to keep the herb constantly moist, add a little urine, and in this situation let it remain eight days longer, continuing to stir it; you afterwards add a pound and a half of pot-ash well pounded, and a pint and a half of stale urine. Let it remain still eight days longer, stirring it as usual; after which you add the same quantity of urine, and at the expiration of five or six days, two drachms of arsenic; it will then be fit for use".

Hellot also quotes Michele (1729, p. 77) in its Latin text. Bancroft (1813, p. 295) gives a résumé of Michele's method, stating that it is "more circumstantial" than that of Imperato and probably "conformable to that of the Florentines". Michele was Keeper of the Grand Duke's gardens in Florence and maybe had direct access to the recipe for preparing the dye. Both processes are virtually identical, except that the sal-ammoniac, sal-gemmae, saltpetre and arsenic are omitted in Michele's method. The time required to complete the latter process is given vaguely as "several weeks" and the fermentation took place in closed wooden vessels. The dye was stored in tight casks, the surface being sprinkled from time to time with urine or lime water, quoted by Hellot as "lye of bone, or gypsum" in the Latin text (Hellot, 1789, p. 233).

With these descriptions in mind, Hellot himself experimented further and made orchil that was fit for use within a fortnight (1789, p. 234). Using half a pound of *Roccella* from Cape Verde and adding fermented urine and slaked lime, in a stoppered glass bottle, he realised that the "volatile urinous spirit", excited by a "terreous alkali", was the means by which the dye was produced. After a fortnight a deep purple dye was produced, though lacking the characteristic smell of violets typical of orchil. At the end of a further three weeks, without additional urine or lime, the "urinous volatile spirit" had evaporated and the substance had acquired a violet smell. It also left the liquor still remaining with the now mainly concentrated substance "a very beautiful crimson". This was the result of reducing the alkalinity of the orchil. Subsequently, by adjusting the alkalinity "blue orchil" and "red orchil" could be manufactured.

This method of preparation, taking about three weeks to a month, seems to have been common usage during the eighteenth and early nineteenth centuries. If the fermentation was allowed to continue beyond the stages when the extraction and formation of the dye were complete, the colour was then spoilt. Therefore, care had to be taken in preserving orchil, which meant that it was generally kept as a soft paste, which was moistened with urine or lime water to keep it in good condition.

Bancroft (1813, p. 381) observed that the purity of the volatile alkali appeared to be important in obtaining the finest-coloured orchil. Probably with this in mind Macintosh utilised ammonia distilled from urine in the preparation of cudbear. This was superseded in the second half of the nineteenth century by the use of ammoniacal by-products of coal-gas extraction. Heat replaced lime in maintaining the fermentation. The methods in use at this time are very clearly explained by Calvert (1876, pp. 231–233). Finally, the role of oxygen in converting orcin into orcein was understood, and by 1918 the lichens were being treated in "large horizontal or vertical cylinders fitted with stirrers and suitable opening for the admission of air" (Perkin and Everest, 1918, p. 558). Also, at this time, orchil was dried and ground into a powder for marketing purposes.

French Purple

The production of French Purple resulting in a fast orchil dye was a more refined process. The method is quoted by Calvert (1876, p. 232) as follows :

"In the first instance, as suggested by Dr. Stenhouse, the lichens were treated with milk of lime, but it was afterwards found to be more advantageous to extract them with a dilute solution of ammonia ; the colour-giving principle is thrown down from the clear solution by hydrochloric acid, and collected on a filter. This precipitate, after being well washed, is dissolved in caustic ammonia, and the ammoniacal liquor kept at an even temperature of 153 to 160 F., for twenty to twenty-five days, when the colour-giving principles of the lichens are transformed into a new series of compounds, the most important of which is a magnificent purple colour. In order to separate this chloride of calcium is added to the liquid, which precipitates it as a fine purple lake. This after being washed and dried is sold under the name of French Purple".

Parelle and Cudbear

The manufacture of parelle and cudbear followed almost the same pattern of development as that of orchil. Hellot made parelle in much the same way as in his experiments with orchil, illustrating that the basis of the recipe was fundamentally similar despite local regional variations in preparation. Alum as well as arsenic were perhaps useless additional chemicals employed at Clermont Ferrand during the eighteenth century (Bancroft, 1813, p. 297). According to Amoreux the preparation of parelle took place in a cool place, such as a cellar or stillroom. Apart from the difference of the lichens used, orchil was also distinguished from parelle and cudbear during the eighteenth and early nineteenth century by being marketed in the form of dry cakes. These were much easier to transport than the paste form of orchil, which had to be kept moist in barrels.

The original recipe for producing cudbear, for which there was an application for a patent in 1758 (see Patent Office Records, 1758, No. 727, Cuthbert Gordon)

includes "spirit of urine", "spirit of soot" (containing ammonia), and quicklime. These were to be "digested" together with the lichens for fourteen days in order to produce cudbear fit for use. It took fourteen days more to dry out and harden. The dry cake form in which it was first sold was superseded in 1813 by the powdered state in which it is still available. By 1876 the method of production of cudbear had become identical to that of orchil, and the final drying process took place in heated troughs fitted with mechanical stirrers to assist the evaporation of the ammonia and water (Calvert, 1876, p. 247).

Because the colours given by the parelle and cudbear lichens are inferior to that of the true *Roccella* orchil, the former dyes seem to have been subject to considerable adulteration in order to improve and vary the colour. The purple produced from *Ochrolechia tartarea* is undoubtedly the best of this group and comes closest to the *Roccella* colour ; many different shades of purple and crimson, some of them quite dull and grey, are obtained from other species. Some species also need longer fermentation to develop the colour. If care was not exercised in the selection of several lichens to make cudbear, the colour could be easily spoiled, though often it seems that miscellaneous collections were used together in this way for the sake of cheapness, and the poor colours resulting were disguised by the addition of brasilwood and magenta.

Variations in colour-content seem also to be due to the conditions, such as habitat, and climate, under which the lichens grew. The colour-producing substances often appear to accumulate in strong sunlight. It is easy to see that *Roccella* species, developing in plenty of sunlight, were considered much more reliable as a source of orchil. It was also a further advantage that, because of their fruticose habit, there was virtually no waste matter such as earth, moss or other lichens to be cleared away before they could be processed.

The most up-to-date method of preparing cudbear is that employed by Messrs. Johnsons of Hendon, Middlesex, who use *Roccella montagnei* from Madagascar. The lichens are not washed or ground up ; they are first boiled in an ammonium carbonate solution, cooled and placed in glazed stoneware sinks about three feet by eighteen inches by twelve inches deep. Ammonia is then added and they are turned and kept in the right state of dampness for three to four weeks. Finally the cudbear is dried out and ground into powder.

Litmus

Litmus has been made in The Netherlands at least since the sixteenth century and its first mention in English is in 1502 (Woodward, 1949, pp. 302–307). It seems to have been manufactured exclusively in The Netherlands until 1940. In the Dictionnaire Universelle de Commerce (1750–65, p. 1621) des Bruslons says that it was a secret known only to the descendants of the inventor, and was made only in Amsterdam. Almost a hundred years later Dr. Ure wrote in his Dictionary of Arts, etc. (1839, p. 786), that the actual process was still guarded as a secret although in theory it must have been understood. Rees' Cyclopoedia (1819), however, describes a process for making litmus but also comments that it could be easily and cheaply obtained from The Netherlands. This suggests that other would-be manufacturers of litmus were unable to compete with the cheap price of the Dutch product.

Litmus is known as turnsole, orseille en pierre, orseille de Flandres, and orseille d'Hollande in France, persis in Germany, and lacmus in Scandinavia.

At first, litmus was probably prepared from the same lichens used for cudbear, chiefly *Ochrolechia tartarea*, but latterly *Roccella* as well as other lichens are used. The oldest method of making litmus resembled that for orchil, with the important difference that extra lime, chalk or gypsum was added to turn the final product blue. The neutralisation of the excess alkalinity, indicated by a change in colour from blue to red, forms the basis of its use in chemical changes involving an alteration in *p*H. Modern and more sophisticated methods of manufacture involve the extraction of red orchil with alcohol to leave a blue residue.

By this modern method of preparation, litmus is not suitable for dyeing textile fibres, but when prepared in the old way it was sometimes used as a dye. Lacmus, the Scandinavian red dye, appears to have been synonymous with the litmus produced by the old method. However, litmus seems not to have been much used by dyers, probably because it did not give as good a colour as orchil. It was sometimes used as a pigment in the painting of miniatures. The chief colouring matter contained in litmus has been called azolitmin; others are erythrolein, erythrolitmin, and spaniolitmin (Calvert, 1876, p. 248).

Another kind of litmus, "tournesole en drapeau", was made in France from a plant of the *Euphorbiaceae*, *La fécule de Ricinoides* Tourn. or *Croton tinctorium* L. It is now called *Chrozophora tinctoria* (see J. Smith, 1882). This litmus was much used by the Dutch for colouring the rinds of cheese.

The Uses of the Orchil Dyes and Methods of Dyeing

As well as the unique quality of the colours given to wool and silk by orchil and related dyes, a great attraction to dyers must have been the ease with which they could be used. "The application of these colours by dyeing is so simple and easy," says Bancroft, "that no instruction can be wanted from me on that subject". The lichen dyes belong to a small class of natural dyestuffs now described as "substantive". This means that no mordant or intermediary agent is required to enable the dye to be taken up by the fibres for which it has an affinity. This seems to have been understood from early times, but sometimes the mordanting substances, acids or alkalies, were used to vary the colour in orchil dyeing.

The Dyeing of Wool and Silk

In general the method of preparing the orchil bath for wool was to add the dye when the water became lukewarm and bring it slowly to the boil. The yarn or cloth to be dyed was then submerged and the bath slowly reheated, the brightest colours being obtained just below boiling point. The cloth or yarn was dyed several times in this way to deepen the colours. The colour from the dye-bath could then be exhausted by boiling for duller and less important colours. Orchil derived from *Roccella* species could withstand boiling during dyeing without dulling the colour, whereas parelle and cudbear deteriorated.

For dyeing silk the bath was heated as before to dissolve the dye. Three or four times the weight of the silk in dye was added if dark shades were needed. The hot liquor was poured off, leaving the residue behind, and the silk was dyed

in this liquor without further heating, being carefully stirred and turned until it reached the required shade. Often the orchil was dissolved and strained off in this way and the liquor used as a stock solution to be added to the dyebaths in specific quantities as required.

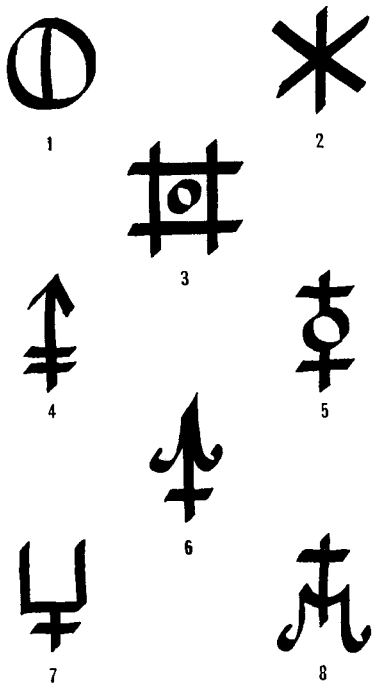

FIG. 1.—Some relative symbols used in lichen dyeing from the old chemical system: 1. Saltpetre. 2. Sal ammoniac. 3. Urine. 4. Iron. 5. Haematite, 6. Arsenic. 7. Potash. 8. Alum. (Redrawn from Koch, R. *The Book of Signs.* Dover Publications Inc., New York.)

The Colours Obtained with Orchil

The natural colour of orchil is a brilliant magenta called in French gris-de-lin and aramante and in English simply purple or red-violet. The addition of soda to the dye-bath will turn this colour to violet, and quicklime and pearl-ash were often used for this purpose. The addition of acid reddens the colour but renders it less brilliant. The Stockholm Papyrus quoted by Faber (1938, p. 290) says that infusions of lemon-leaves, barley, penny-wort or onion juice were used to improve the fastness of orchil. Dambourney (1794, p. 167) explains a method of passing woollen cloth through a bath of vinegar or cold water and oil of vitriol after dyeing with orchil, which he thought improved the fastness. This treatment turned the colour a lively red, but immersion in a soap bath evidently restored some of the blue and left the cloth a "charming rose colour".

Alum mordant (aluminium potassium sulphate) has much the same effect as acids on the final colour, although the effect seems to me a little brighter. Berthollet found, however, that alum with orchil spoiled the lustre, and quite destroyed the colour of parelle. There may have been impurities in his unrefined alum which caused this effect. From personal experience the colour obtained with alum and *Roccella* orchil is rather clear and bright. Iron mordant (ferrous sulphate) deepens the original colour to a rich reddish purple. Iron was used as a mordant in various

forms in ancient times and haematite (Fe_2O_3), chalkanthon and ferro-acetate are mentioned in the Stockholm Papyrus (see Born, 1937, p. 113). If the stuff dyed in orchil and iron was then dipped in a solution of tannin, the colour became even deeper. Tannin was obtained from oak-galls and sumach* and it occurs in all woods that give brown and black dyes with iron. Orchil was widely used in brown dyeing, particularly in the nineteenth century.

A tin mordant is the only one which makes the orchil dyes appreciably faster. Today stannous chloride, often with the addition of oxalic acid, are the ingredients of tin mordant. Before purified mordants were obtainable the "composition for scarlet" consisted of granulated metallic tin dissolved in a weak aqua regia composed of river water, spirit of nitre, sal ammoniac and saltpetre (Hellot, 1789, p. 122). This composition was also the means by which cochineal was made to dye scarlet. It was also used with orchil to make it faster, but all the writers lament the change of colour. This was a red colour known as "half scarlet", but was much nearer to crimson according to Bancroft and is from my own experience quite deep but rather dull. When using orchil as a ground or final dip with other dyes, the effect of the mordants used with the latter upon the orchil colour had to be borne in mind.

The Orchil Dyes Used with other Dyestuffs in Compound Colours

Before the industrial revolution dyers were divided into several categories which appear to change over the years and according to the individual development of dyes and their techniques. In earliest times and up to the sixteenth century, all these categories tended to be related to colour. One became qualified to dye in purple, blue, black and so on, and was occupied chiefly with one colour and often one basic dyestuff. The murex purple and woad blue dyers are examples of this; dyestuffs of other colours entered into their processes, but purely in relation to the variations of colour and quality of purple and blue that were specifically required by them. Gradually, as dyeing increased as a business and a greater variety of dyestuffs became available a more important division became that between the dyers of "true colours" (i.e. fast)†, "grand teint" or "bon teint" in French, and the dyers of "false" colours ("faux teint") (Hellot *et al.*, 1787, p. vi)‡. It follows that the basic dyestuffs of the high-class colours were much more expensive and less readily available than those of the second class. Indigo and cochineal had to be imported from the East and West Indies; madder, kermes and woad were available in limited quantities from European sources, but to meet increasing demand these also had to be imported from the East. Again, other dyestuffs were used with these fast colours, but the amount and manner in which this was permitted was very severely controlled by law. (See Amoreux, 1787, p. 81.)

It was accepted that the orchil dyes were only used by the second class of dyers producing cheaper goods, but the dyers of "true colours" were increasingly tempted

*A preparation of the dried and chopped leaves and shoots of *Rhus*, especially *Rhus coriaria* much used in tanning.

†Woad and indigo for blues; madder and cochineal for reds; weld (*Reseda luteola*) and dyer's greenwood (*Genista tinctoria*) for yellows; these were the most important fast dyes for silk and wool.

‡An exception to this appears to have been that scarlet dyeing remained as a separate category until about the mid-eighteenth century.

to use them. "It (orchil) even has the effect", says Amoureux, "of bringing out the finest and best colours in these dyes and of making them go further; two inestimable advantages". But as there was no forthcoming improvement in the fastness of orchil in combination with other colours, it was only to be used judiciously. For instance, it was not possible to make a deep blue on silk with indigo, but an orchil ground was found to be an excellent way of achieving this, although it produced a rather redder blue than the pure dark indigo. In dyeing blue with woad in ancient times, the orchil seems either to have been added to the blue vat or given as a separate finish to the stuff after blueing (Faber, 1938, p. 289). But in the eighteenth century both silk and wool were given the ground of orchil and finished with the woad or indigo (Amoreux, 1787, pp. 76–77). English blue broad-cloths, for which the woollen mills of Gloucestershire and Norwich were famous in the seventeenth and eighteenth centuries, were dyed in this way (Vinçard, 1820, p. 125).

In madder dyeing the purpose of orchil was to "blue" the colour, turning the naturally yellowish madder-red to crimson. In this case the stuff was dyed with orchil after maddering. Red cloth for French cavalry and infantry uniforms were dyed in this way during the eighteenth century (Amoreux, 1787, p. 77). In Thomas Packer's "The Dyer's Guide" (1816, p. 58) there is a recipe for dyeing wool maroon with madder, peach wood* and orchil. Madder and orchil also appear in brown dyeing with a number of other ingredients. An early reference to this comes in the Statute of Henry VIII of 1533, which "ordered that none should dye woollen cloth 'as brown blewes, pewkes†, tawnayes, or vyolettes' unless they were 'perfectly boyled, greyned (cochineal) or madered upon the wode and shotte with good and sufficient corke or orchall' ". (See Salzmann, 1923, p. 212.) Some interesting recipes are also to be found in a dyer's pattern book of wools used in the carpet factory of David Hindley at Liversage, Yorkshire, 1853–1878 (V. & A. Textile Dept. T.8–1943). A full red-brown was made with redwood*, fustic chips‡, cudbear and argol (tartar); pale pinkish fawn, called brown, with argol, fustic chips‡, madder, redwood, cudbear and copperas; "oak", a very pink light brown, with fustic, madder cudbear and copperas. A softer, warm brown called "drab" was made with madder, fustic, cudbear and argol; another drab, a very pale off-white, with cudbear and fustic; and a third, mauvish-grey, with alum, logwood (*Haematoxylon campechianum*) and cudbear. There is no indication of the method of dyeing, but in all the brown recipes except one cudbear appears as the last-named dyestuff. There are also two recipes and patterns of crimson dyed with cudbear and urine.

Crimson and pinkish reds were mostly dyed with kermes or cochineal by the good dyers; the fugitive colours being derived from root of alkanet (*Anchusa sempervirens*), carthamus flowers (safflower, *Carthamus tinctoria*), annotto§ and brasil woods. The Stockholm Papyrus gives a recipe for a rose colour made with kermes and "common orchil" over a woad ground (Faber, 1938, p. 290). Packer (1816)

*Brasil woods—*Caesalpinia crista* L. and other leguminous trees (sappan, lima, peachwood).

†Usually a brown colour made with galls and copperas, between russet and black; sometimes a deep dark purple was called pewke or puke (puce) (Oxford Eng. Dict. 1933).

‡The name of two kinds of wood, *Rhus cotinus* (Sumach) and *Cladrastis lutea*, both used for dyeing yellow.

§Or Anatta, an orange-red dye from the pulp surrounding the seeds of *Bixa orellana* often used to colour cheese.

has a recipe for worsted yarn dyed crimson with cochineal and finished with orchil. The yarn was mordanted first with 2½ oz. alum and 1½ oz. tartar to the pound and then dyed in cochineal; it was finished in an orchil bath prepared as for silk. The lightest and brightest colours were to be dyed first, the deepest second, and "common compound colours" last. Silk was also dyed crimson in this way. Other reds, between crimson and scarlet, were obtained with orchil and the "composition for scarlet". In this case silk could be prepared with the tin composition as a mordant and the orchil added to the bath with the other dyes (Berthollet & Berthollet, 1841, p. 369). Tin mordanting is generally done at the end of the process in dyeing wools.

In the dyeing of purple and lilac colours the Stockholm Papyrus gives probably the clearest of early descriptions which include orchil. W. Born (1937, p. 113) quoting from this Papyrus says that there was a cold and a hot method of making the counterfeit purple. The cold method seems to have involved using red dyes on a blue ground; in the hot method red and purplish dyes were combined with iron, copper and galls to deepen the colour. Orchil was used in both methods. Descriptions of the use of orchil in connection with murex purple by the Phoenicians, Greeks and Romans have been included in the first part of this paper.

Delormois (1726, p. 414), much of whose information apparently derives from Roseto, has three recipes for dyeing silk stockings with orchil. For purple stockings, orchil alone is used; for violet, orchil and "a little handful of quicklime"; and for lavender ½ oz. orchil, 1 oz. of powdered galls, 1 oz. copperas were required to dye two pairs of stockings. The ingredients were dissolved in the dye-bath, which was brought to the boil and stirred, and the stockings were then put in.

Macquer's chapters (see Hellot *et al.*, 1789, pp. 335–348) on the various shades of "fine" and "false" violets on silk are very interesting. He says that the "blue of all violets" is obtained from indigo; when reddened with cochineal it becomes fine violet; reddened by other means, particularly orchil, it is a "false violet". However, orchil violets are often described as the finest of the false violets and the most frequently in use. The silk was first dyed in orchil to a suitable depth and then dipped in the indigo vat. If the colour was not dark enough the process was repeated. The deepest colours were dyed in this way and called "violet of Holland". A redder colour, with a full orchil ground but lighter blue finish, was called "bishop's violet". A specially weak indigo vat had to be made for dyeing lilac, with some pearl ash (potassium carbonate) added to help in blueing the orchil. For dark shades of violet the silk was sometimes dyed first with brasil wood, then with orchil and finally in the indigo vat. This was evidently cheaper and a saving of orchil, but the colour was not so good, as brasil wood required an alum mordant which spoiled the brightness of the orchil. After dyeing, the silk had to be dried under cover to prevent the colour changing in the air.

In dyeing fine violet colours, the orchil was added to the cochineal bath when the colour from this had been exhausted. Macquer denounced this practice as "a shameful abuse", at the same time admitting that for pale violet colours orchil was a necessity, as in light shades cochineal would only give unpleasant and dull results. Packer's (1816) recipes for dyeing violets on silk are taken from Macquer (see Hellot *et al.*, 1789).

Sir J. E. Smith (see note in Herb., sheet 1693.9) notes that cudbear was used at Norwich and in Gloucestershire in the dyeing of all crimson and purple shades.

The names of some of the purple shades are given by Berthollet & Berthollet (1841, p. 368) as "violets, lilacs, mallows and rosemary flowers"; "soupe au vin", a deep colour made by adding an alkali; "agates, rosemary flowers and other delicate shades" also made with alkali.

Three shades of purple are shown in the Hindley pattern book (1853–78). The first is a fairly deep colour dyed with twenty pounds of cudbear and two and a half pounds of alum to "3 score" of wool. The second colour, brighter and not so deep, is dyed in the same bath after the first colour and reddened with "Tartar, Alum or Argol". The third colour is mauve and dyed after the second with no addition to the bath. A pale greyish lilac is made with three and a half pounds of cudbear, six pounds of logwood and one and a half pounds of alum to forty pounds of wool.

Orchil Colours in Textile Printing

In the printing of wool and silk, orchil was used to obtain a dark blue as an addition to logwood and for a dark aramanth with cochineal. The method was to mix the dyes together with their mordants and a starch paste thickener. Nitrate of iron was the mordant used for the blue and a small amount of soluble Prussian blue (hydrated ferric ferrocyanide generally mixed with varying quantity of potassiferrous ferricyanide) was added. Alum and perchloride of tin were the mordants for the aramanth colour. After printing, the cloth was steamed to fix the dye. It received no preparation prior to printing except for bleaching and removing the gum from silk where necessary (see Ure, 1839, p. 236).

For dyeing cotton with French purple, the cloth was evidently first mordanted with albumen, or prepared as for the Turkey-red style of calico printing (Calvert, 1876, p. 232). By the latter method the pattern was printed on to the prepared cloth with various mordants. The cloth was then dyed in a madder bath and the different mordants produced various colours—black, brown, fawn, red and purple shades. The madder dye did not fix permanently to the unmordanted background; nevertheless the process of clearing this back to white was quite tiresome. If blue, green and yellow were also required in the pattern, these colours had to be added afterwards. Indigo "pencilling" provided the blue, and iron buff (ferrous acetate fixed in an alkaline bath) was a yellow frequently employed. Green was a combination of blue and yellow dyes.

The preparation of the cloth before the printing and dyeing could take place was itself a lengthy process. As cotton has no natural affinity for madder, this had to be created by means of the mordants, and these in themselves are not sufficient to make the dye fast on cotton as they do on silk and wool. The cotton was thus prepared with oil, which created suitable conditions for the mordants to act. The method was to steep the cloth in olive oil, or castor oil, dry it under heat and then pass it through a solution of pearl ash. This process was repeated from six to eight times where very intense and permanent colours were required, probably several days' work. However, by the time French purple came into use, a preparation of soluble oils, called "Turkey red oil", was available which enabled the oiling of the cloth to be completed in one operation. The only technical reference to the use of French purple I have so far discovered is that of Calvert in his "Dyeing and Calico Printing" (1876, pp. 231–232). It is not clear from this account what

happened after the cloth had been prepared for printing, but it appears that French purple was used in calico printing to produce an all-over colour rather than individuality as part of a multi-coloured design. Chambers Encyclopaedia (Anon. (3), 1908, pp. 389–390) mentions French purple in connection with calico printing and states that it was soon superseded by mauveine, the first coal-tar dye. Lindsay says that French purple competed favourably with synthetic dyes in 1867 (1867, pp. 101–109).

Orchil Dyes in the Present Century

In 1908 Chambers Encyclopaedia (Anon. (3)) stated that Archil was "much employed in the dyeing of silks" and Cudbear "very useful to the wool dyer". In 1914, according to Alfred Edge (1914, pp. 186–188) "orchil and cudbear still find large employment in woollen dyeing". As recently as 1962 the Encyclopaedia Britannica stated that "archil is chiefly employed for the dyeing of carpet yarns". A possible explanation for its continuing in use so late in the carpet industry is that in the pile of a rug or carpet the colour does not fade so quickly as on the surface of a woven cloth. Where the colour is exposed at the end of the pile fading occurs, but strong colour remains underneath. As the top of the pile is gradually being worn away the stronger colour is continually coming to the top, and with the movement of the pile this stronger colour does in fact show from time to time and affects the total colouring of the carpet.

Regarding the fading of orchil dyes, from my own observations I find that in four weeks wools left in a north light have lost their vivacity; in about a year, if exposed all the time, most of the colour has drained away. But wools kept in boxes or drawers and brought out from time to time for demonstration have remained almost as bright as when first dyed for two or three years and even longer. This may explain why some of the early Scottish tartans that have survived and are said to have been dyed with lichens have retained so much of their red colours. It may be that these were garments kept for special wear and worn only occasionally and otherwise closeted away in the dark.

Instructions for dyeing with orchil have appeared in some publications of the present century intended for the use of hand-craftsmen. Ethel Mairet's "A Book on Vegetable Dyes" (1938) and Violetta Thurston's "The Use of Vegetable Dyes" (1957) are two of the most important of these publications. More recently, Miss Eileen Bolton's "Lichens for Vegetable Dyeing" (1960, pp. 44–49) has a very informative section on the making of orchil from several different species, and she gives a colour chart of the shades produced.

In considering the data available in connection with a study of this nature, it has become clear that it is important to distinguish between the dyes used in antiquity and those in production from the late seventeenth century onwards. In the former, reference is limited to interpretation of the texts of authorities, whereas in the latter period there are also textiles, account books, samples, order books and treatises to hand. It seems clear that fresh research is needed in the latter field, incorporating data from these documents and chronicles of the fourteenth century onwards. I am aware of no experiments being performed in this country to detect the presence and nature of lichen dyes in actual samples of cloth. Perhaps this paper will encourage more research along these lines.

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